In [1]:
```
import numpy as np
```

In [2]:
```
np.__version__
```

Out[2]:
```
'1.26.4'
```

In [3]:
```
my_list = [1,2,3,4,5,6,7]
my_list
```

Out[3]:
```
[1, 2, 3, 4, 5, 6, 7]
```

In [4]:
```
type(my_list)
```

Out[4]:
```
list
```

In [5]:
```
len(my_list)
```

Out[5]:
```
7
```

In [6]:
```
arr = np.array(my_list)
arr
```

Out[6]:
```
array([1, 2, 3, 4, 5, 6, 7])
```

In [7]:
```
type(arr)
```

Out[7]:
```
numpy.ndarray
```

arange

In [8]:
```
np.arange(10)  # arange --- it will print the value within intervals
```

Out[8]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [9]:
```
np.arange(10,15)
```

Out[9]:
```
array([10, 11, 12, 13, 14])
```

In [10]:
```
np.arange(10,20,5)  # in array only three argument passing is allowed
```

Out[10]:
```
array([10, 15])
```

In [11]:
```
np.arange(10,20,5,3)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[11], line 1
----> 1 np.arange(10,20,5,3)

TypeError: Cannot interpret '3' as a data type
```

In [22]:
```
np.arange(10,20)
```

Out[22]:
```
array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

In [23]:
```
np.arange(20,10)   # 1st argument is always have to grater than second argument
```

Out[23]:
```
array([], dtype=int32)
```

In [24]:
```
np.arange(-20,10)
```

Out[24]:
```
array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10,  -9,  -8,
        -7,  -6,  -5,  -4,  -3,  -2,  -1,   0,   1,   2,   3,   4,   5,
         6,   7,   8,   9])
```

zeros

In [12]:
```
np.zeros(5)  # parameter tuning
```

Out[12]:
```
array([0., 0., 0., 0., 0.])
```

In [13]:
```
np.zeros(5, dtype=int) # hyperparameter tuning
```

Out[13]:
```
array([0, 0, 0, 0, 0])
```

In [15]:
```
np.zeros([2,2],dtype=int) # 2d array
```

Out[15]:
```
array([[0, 0],
       [0, 0]])
```

In [16]:
```
np.zeros([5,10],dtype=int)
```

Out[16]:
```
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

ones

In [19]:
```
np.ones(3)  # it will give one as per mentioned arguments   # parameter tuning
```

Out[19]:
```
array([1., 1., 1.])
```

In [20]:
```
np.ones([5,10],dtype=int) # hyperparameter tuning
```

Out[20]:
```
array([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])
```

In [21]:
```
np.ones([3,5])
```

Out[21]:
```
array([[1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.]])
```

twos

In [25]:
```
np.twos(3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[25], line 1
----> 1 np.twos(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:333, in __getattr__(attr)
    330     "Removed in NumPy 1.25.0"
    331     raise RuntimeError("Tester was removed in NumPy 1.25.")
--> 333 raise AttributeError("module {!r} has no attribute "
    334                      "{!r}".format(__name__, attr))

AttributeError: module 'numpy' has no attribute 'twos'
```

In [26]:
```
np.threes(3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[26], line 1
----> 1 np.threes(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:333, in __getattr__(attr)
    330     "Removed in NumPy 1.25.0"
    331     raise RuntimeError("Tester was removed in NumPy 1.25.")
--> 333 raise AttributeError("module {!r} has no attribute "
    334                      "{!r}".format(__name__, attr))

AttributeError: module 'numpy' has no attribute 'threes'
```

random.rand

In [28]:
```
random.rand(2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[28], line 1
----> 1 random.rand(2)

NameError: name 'random' is not defined
```

In [30]:
```
np.random.rand(2)# it will always give differnt or random values
```

Out[30]:
```
array([0.42267845, 0.52889848])
```

In [32]:
```
np.random.rand(5)
```

Out[32]:
```
array([0.838171  , 0.92163989, 0.08441742, 0.11428107, 0.51959533])
```

In [34]:
```
np.random.rand(3,2)
```

Out[34]:
```
array([[0.25909326, 0.76722082],
       [0.29060015, 0.6423524 ],
       [0.68566679, 0.65493707]])
```

random.randint

In [37]:
```
np.random.randint(3) # it also give random values but in integer
```

Out[37]:
```
2
```

In [38]:
```
np.random.randint(5,10)# it will give values between 5-9 randomly
```

Out[38]:
```
8
```

In [39]:
```
np.random.randint(5,10,3)
```

Out[39]:
```
array([9, 8, 5])
```

In [40]:
```
ran = np.random.randint(5,10,3)
print(ran)
```

```
[6 7 5]
```

In [41]:
```
otp = np.random.randint(5,10,4)
print(otp)
```

```
[7 7 5 7]
```

In [42]:
```
arr0 = np.random.randint(10,40,(10,10))# first 10 is rows and 2nd 10 is column
print(arr0)   # first 10 and 40 is a range  values between 10-40
```

```
[[16 24 33 16 16 25 24 36 21 34]
 [14 15 13 10 21 18 36 20 19 31]
 [18 33 21 25 21 23 20 23 28 25]
 [31 33 29 23 38 12 32 14 37 12]
 [26 27 34 18 11 23 30 14 25 20]
 [23 12 37 32 11 15 18 12 18 16]
 [22 36 28 26 10 29 25 14 24 12]
 [18 37 20 14 37 17 12 17 34 28]
 [29 29 28 17 35 24 25 36 35 19]
 [14 33 36 10 17 24 31 16 16 35]]
```

In [43]:
```
arr0
```

Out[43]:
```
array([[16, 24, 33, 16, 16, 25, 24, 36, 21, 34],
       [14, 15, 13, 10, 21, 18, 36, 20, 19, 31],
       [18, 33, 21, 25, 21, 23, 20, 23, 28, 25],
       [31, 33, 29, 23, 38, 12, 32, 14, 37, 12],
       [26, 27, 34, 18, 11, 23, 30, 14, 25, 20],
       [23, 12, 37, 32, 11, 15, 18, 12, 18, 16],
       [22, 36, 28, 26, 10, 29, 25, 14, 24, 12],
       [18, 37, 20, 14, 37, 17, 12, 17, 34, 28],
       [29, 29, 28, 17, 35, 24, 25, 36, 35, 19],
       [14, 33, 36, 10, 17, 24, 31, 16, 16, 35]])
```

In [44]:
```
arr1 = np.random.randint(1,5,(3,3))
print(arr1)
```

```
[[1 4 1]
 [2 4 3]
 [2 4 1]]
```

indexing and slicing in array

In [45]:
```
arr1[:]
```

Out[45]:
```
array([[1, 4, 1],
       [2, 4, 3],
       [2, 4, 1]])
```

In [46]:
```
len(arr1)
```

Out[46]:
```
3
```

In [47]:
```
arr1.shape  # 1st 3 is count of rows and 2nd 3 is a count of columns
```

Out[47]:
```
(3, 3)
```

In [48]:
```
arr0.shape
```

Out[48]:
```
(10, 10)
```

In [49]:
```
len(arr0)
```

Out[49]:
```
10
```

In [50]:
```
arr1
```

Out[50]:
```
array([[1, 4, 1],
       [2, 4, 3],
       [2, 4, 1]])
```

In [51]:
```
arr1[0]  # print row
```

Out[51]:
```
array([1, 4, 1])
```

In [52]:
```
arr1[1] # print 1st row
```

Out[52]:
```
array([2, 4, 3])
```

In [53]:
```
arr1
```

Out[53]:
```
array([[1, 4, 1],
       [2, 4, 3],
       [2, 4, 1]])
```

In [54]:
```
arr1[:,1]  # print 1st column
```

Out[54]:
```
array([4, 4, 4])
```

In [55]:
```
arr1[:,0] # print 0 column
```

Out[55]:
```
array([1, 2, 2])
```

In [56]:
```
arr1[:,2] # 3rd column
```

Out[56]:
```
array([1, 3, 1])
```

In [57]:
```
arr1
```

Out[57]:
```
array([[1, 4, 1],
       [2, 4, 3],
       [2, 4, 1]])
```

In [58]:
```
arr1[0]
```

Out[58]:
```
array([1, 4, 1])
```

In [60]:
```
v = arr1[1,2]  # 1st row and 2nd column element
print(v)
```

```
3
```

In [61]:
```
v = arr1[1,0]  # 1st row and 0 column element
print(v)
```

```
2
```

In [62]:
```
arr0
```

Out[62]:
```
array([[16, 24, 33, 16, 16, 25, 24, 36, 21, 34],
       [14, 15, 13, 10, 21, 18, 36, 20, 19, 31],
       [18, 33, 21, 25, 21, 23, 20, 23, 28, 25],
       [31, 33, 29, 23, 38, 12, 32, 14, 37, 12],
       [26, 27, 34, 18, 11, 23, 30, 14, 25, 20],
       [23, 12, 37, 32, 11, 15, 18, 12, 18, 16],
       [22, 36, 28, 26, 10, 29, 25, 14, 24, 12],
       [18, 37, 20, 14, 37, 17, 12, 17, 34, 28],
       [29, 29, 28, 17, 35, 24, 25, 36, 35, 19],
       [14, 33, 36, 10, 17, 24, 31, 16, 16, 35]])
```

In [63]:
```
arr0[-7]  # print -7 row
```

Out[63]:
```
array([31, 33, 29, 23, 38, 12, 32, 14, 37, 12])
```

In [64]:
```
arr0[:,-7]  # it will print -7 column
```

Out[64]:
```
array([16, 10, 25, 23, 18, 32, 26, 14, 17, 10])
```

In [65]:
```
arr0
```

Out[65]:
```
array([[16, 24, 33, 16, 16, 25, 24, 36, 21, 34],
       [14, 15, 13, 10, 21, 18, 36, 20, 19, 31],
       [18, 33, 21, 25, 21, 23, 20, 23, 28, 25],
       [31, 33, 29, 23, 38, 12, 32, 14, 37, 12],
       [26, 27, 34, 18, 11, 23, 30, 14, 25, 20],
       [23, 12, 37, 32, 11, 15, 18, 12, 18, 16],
       [22, 36, 28, 26, 10, 29, 25, 14, 24, 12],
       [18, 37, 20, 14, 37, 17, 12, 17, 34, 28],
       [29, 29, 28, 17, 35, 24, 25, 36, 35, 19],
       [14, 33, 36, 10, 17, 24, 31, 16, 16, 35]])
```

In [66]:
```
arr0[3:4, 2:6]  # it will print  in 3 rd row and 2-6 column element
```

Out[66]:
```
array([[29, 23, 38, 12]])
```

In [67]:
```
arr0[4:5, 3:7]  # it will firstly select row and than looking for columns in bet
```

Out[67]:
```
array([[18, 11, 23, 30]])
```

In [69]:
```
arr0
```

Out[69]:
```
array([[16, 24, 33, 16, 16, 25, 24, 36, 21, 34],
       [14, 15, 13, 10, 21, 18, 36, 20, 19, 31],
       [18, 33, 21, 25, 21, 23, 20, 23, 28, 25],
       [31, 33, 29, 23, 38, 12, 32, 14, 37, 12],
       [26, 27, 34, 18, 11, 23, 30, 14, 25, 20],
       [23, 12, 37, 32, 11, 15, 18, 12, 18, 16],
       [22, 36, 28, 26, 10, 29, 25, 14, 24, 12],
       [18, 37, 20, 14, 37, 17, 12, 17, 34, 28],
       [29, 29, 28, 17, 35, 24, 25, 36, 35, 19],
       [14, 33, 36, 10, 17, 24, 31, 16, 16, 35]])
```

In [68]:
```
arr0[6:8, 2:6]   # it will print element in 6 and 7 row between 2-6 columns rang
```

Out[68]:
```
array([[28, 26, 10, 29],
       [20, 14, 37, 17]])
```

In [70]:
```
arr0[3:6, 4:9]
```

Out[70]:
```
array([[38, 12, 32, 14, 37],
       [11, 23, 30, 14, 25],
       [11, 15, 18, 12, 18]])
```

In [71]:
```
arr1
```

Out[71]:
```
array([[1, 4, 1],
       [2, 4, 3],
       [2, 4, 1]])
```

In [72]:
```
arr1[1]
```

Out[72]:
```
array([2, 4, 3])
```

In [75]:
```
arr1[1,2] # 1st row and 2 means column
```

Out[75]:
```
3
```

In [76]:
```
arr1
```

Out[76]:
```
array([[1, 4, 1],
       [2, 4, 3],
       [2, 4, 1]])
```

In [77]:
```
arr1[1:2]
```

Out[77]:
```
array([[2, 4, 3]])
```

In [79]:
```
arr1[0:1]  # it will print row
```

Out[79]:
```
array([[1, 4, 1]])
```

In [ ]:

```

```